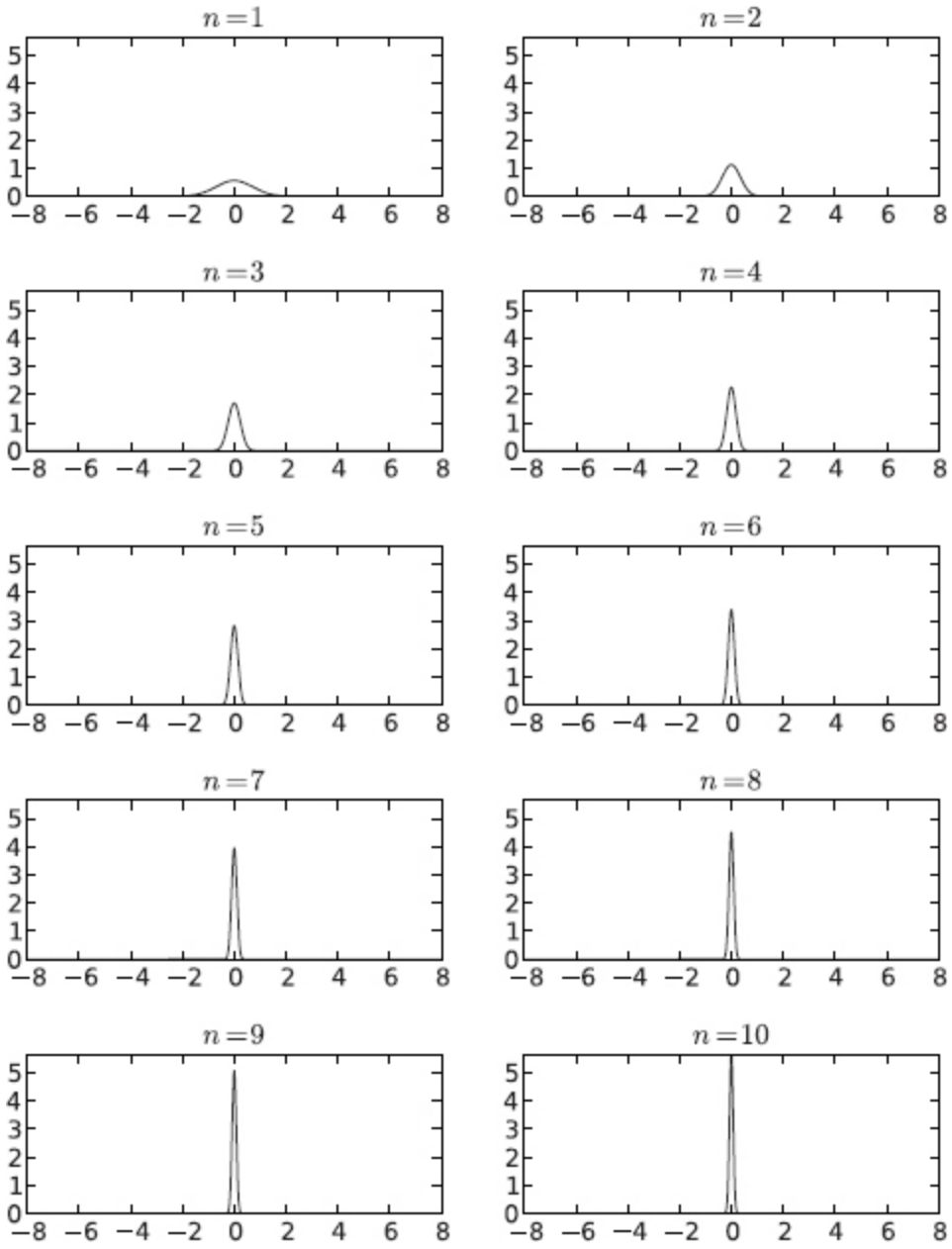


Figure 8.1: Dirac Delta Function Approximations. These approximations are based on $\frac{n}{\sqrt{\pi}}e^{-(nx)^2}$ and plotted for increasing values of *n*.

8.1.3 Integration by Parts

Before discussing linear operators, we'll take a short detour to remind you of a technique called *integration by parts*. It's fairly simple, and indispensable for our purposes. We'll be using it again and again. Suppose